

Name: _____ Period: _____



Unit 6: Electrochemistry

AP Classroom Partial Topic 9

Unit Map

Textbook Reference: Zumdahl *Chemistry* Chapter 4, Chapter 18

Date	Day	Learning Targets	Activities	Work on this
1/6-7	U6D1	What is a redox reaction?	- Review- what are oxidation numbers and what is redox? - Oxidation Reduction POGIL	✓ Oxidation Reduction POGIL ✓ Watch the Ecell EdPuzzle
1/8-9	U6D2	What is a redox titration??	- Review POGIL - Balancing redox rxns - Redox Titration and the return of Dr. Jay Patel	✓ Textbook Prob Set 1 ✓ Work on Redox Titration
1/12-13	U6D3	What is a galvanic cell?	Notes and Practice: Electrochemical Cells and calculating cell potential	✓ Textbook Prob Set 1 ✓ Finish Redox Titration ✓ Textbook Prob Set 2?
1/14-15	U6D4	How do we calculate cell potential	Notes and Practice: Calculating cell potential	✓ Quiz next time! ✓ Finish Prob Set 1 ✓ Textbook Prob Set 2?
1/16-20	U6D5	What are nonstandard conditions?	- Quiz - Notes and Practice: Spontaneity and electrochemical cells	✓ Textbook Prob Set 2 ✓
1/21-22	U6D6	What is an electrolytic cell?	- Lab - Electrolytic Cells - Review	✓ Study for exam ✓ Finish Prob Set 2
1/23-26	U6D7	Unit 6 Exam		

Unit 6: Electrochemistry Learning Targets (CED Topics 4, 9)

Unit 6 Electrochemistry	
4.9	Oxidation Reduction Reactions
4.9A	Represent a balanced redox reaction equation using half-reactions.
	4.9.A.1: Balanced chemical equations for redox reactions can be constructed from half-reactions.
9.8	Galvanic (Voltaic) and Electrolytic Cells
9.8A	Explain the relationship between the physical components of an electrochemical cell and the overall operational principles of the cell.
	9.8.A.1: Each component of an electrochemical cell (electrodes, solutions in the half-cells, salt bridge, voltage / current measuring device) plays a specific role in the overall functioning of the cell. The operational characteristics of the cell (galvanic vs. electrolytic, direction of electron flow, reactions occurring in each half-cell, change in electrode mass, evolution of a gas at an electrode, ion flow through the salt bridge) can be described at both the macroscopic and particulate levels.
	9.8.A.2: Galvanic, sometimes called voltaic, cells involve a thermodynamically favored reaction, whereas electrolytic cells involve a thermodynamically unfavored reaction. Visual representations of galvanic and electrolytic cells are tools of analysis to identify where half-reactions occur and in what direction current flows.
	9.8.A.3: For all electrochemical cells, oxidation occurs at the anode and reduction occurs at the cathode. ★
9.9	Cell Potential and Free Energy
9.9A	Explain whether an electrochemical cell is thermodynamically favored, based on its standard cell potential and the constituent half-reactions within the cell.
	9.9.A.1: Electrochemistry encompasses the study of redox reactions that occur within electrochemical cells. The reactions are either thermodynamically favored (resulting in a positive voltage) or thermodynamically unfavored (resulting in a negative voltage and requiring an externally applied potential for the reaction to proceed).
	9.9.A.2: The standard cell potential of electrochemical cells can be calculated by identifying the oxidation and reduction half-reactions and their respective standard reduction potentials.
	9.9.A.3: ΔG° (standard Gibbs free energy change) is proportional to the negative of the cell potential for the redox reaction from which it is constructed. Thus, a cell with a positive E° involves a thermodynamically favored reaction, and a cell with a negative E° involves a thermodynamically unfavored reaction. EQN: $\Delta G^\circ = -nFE^\circ$
9.10	Cell Potential under Non-standard Conditions
9.10A	Explain the relationship between deviations from standard cell conditions and changes in the cell potential.
	9.10.A.1: In a real system under nonstandard conditions, the cell potential will vary depending on the concentrations of the active species. The cell potential is a driving force toward equilibrium; the farther the reaction is from equilibrium, the greater the magnitude of the cell potential.
	9.10.A.2: Equilibrium arguments such as Le Châtelier's principle do not apply to electrochemical systems, because the systems are not in equilibrium.
	9.10.A.3: The standard cell potential E° corresponds to the standard conditions of $Q = 1$. As the system approaches equilibrium, the magnitude (i.e., absolute value) of the cell potential decreases, reaching zero at equilibrium (when $Q = K$). Deviations from standard conditions that take the cell further from equilibrium than $Q = 1$ will increase the magnitude of the cell potential relative to E° . Deviations from standard conditions that take the cell closer to equilibrium than $Q = 1$ will decrease the magnitude of the cell potential relative to E° . In concentration cells, the direction of spontaneous electron flow can be determined by considering the direction needed to reach equilibrium.
	9.10.A.4: Algorithmic calculations using the Nernst equation are insufficient to demonstrate an understanding of electrochemical cells under nonstandard conditions. However, students should qualitatively understand the effects of concentration on cell potential and use conceptual reasoning, including the qualitative use of the Nernst equation to solve problems: EQN: $E = E^\circ - (RT/nF) \ln Q$
9.11	Electrolysis and Faraday's Law
9.11A	Calculate the amount of charge flow based on changes in the amounts of reactants and products in an electrochemical cell.
	9.11.A.1: Faraday's laws can be used to determine the stoichiometry of the redox reaction occurring in an electrochemical cell with respect to the following: Number of electrons transferred, . Mass of material deposited on or removed from an electrode (as in electroplating), current, time elapsed, charge of ionic species. EQN: $I = q/t$

Section 1: REDOX Reactions and Assigning Oxidation States

- A reaction in which _____ are transferred from one atom to another is called an oxidation and reduction reaction or REDOX reaction.
- Oxidation numbers are used to keep track of how many electrons are _____ or _____ by each atom.
- These oxidation numbers are assigned according to a set of rules.
- Oxidation numbers are very similar to _____, but they are not just found in _____ substances, but also _____.
- The oxidation number assigned to an atom in a molecule is based on the distribution of electrons in that molecule.
- These oxidation numbers are assigned according to a set of rules.

Rules for applying Oxidation Numbers

Rule	Example
1. The oxidation number of any uncombined element is 0.	The oxidation number of Na(s) is 0.
2. The oxidation number of a monatomic ion equals the charge on the ion.	The oxidation number of Cl^- is -1.
3. The more-electronegative element in a binary compound is assigned the number equal to the charge it would have if it were an ion.	The oxidation number of O in NO is -2.
4. The oxidation number of fluorine in a compound is always -1.	The oxidation number of F in LiF is -1.
5. Oxygen has an oxidation number of -2 unless it is combined with F, in which it is +1 or +2, or it is in a peroxide, in which it is -1.	The oxidation number of O in NO_2 is -2.
6. Hydrogen's oxidation state in most of its compounds is +1 unless it is combined with a metal, in which case it is -1.	The oxidation number of H in LiH is -1.
7. In compounds, the elements of Groups 1 and 2 as well as aluminum have oxidation numbers of +1, +2, and +3, respectively.	The oxidation number of Ca in CaCO_3 is +2.
8. The sum of the oxidation numbers of all atoms in a neutral compound is 0.	The oxidation number of C in CaCO_3 is +4.
9. The sum of the oxidation numbers of all atoms in a polyatomic ion equals the charge of the ion.	The oxidation number of P in H_2PO_4^- is +5.

Practice:

Give the oxidation number for the following:

Assigning Oxidation Numbers in Compounds: Find the oxidation numbers for all of the atoms in HNO_3 .

Strategy: Do the easy things first! Look for the bosses!

Rule #5: O _____ Rule #6: H _____

Then work on the less predictable parts.

Rule #8: The overall charge for the neutral compound is zero.

H N O₃

$$(+1) + (\text{Charge of N}) + 3(-2) = 0$$

Oxidation State of N: _____

More Practice:

- N_2O_3
- NO_3^{-1}
- AgNO_3
- H_2SO_4
- $\text{Fe}(\text{ClO}_2)_3$

Oxidation Numbers Practice

Directions: Use the *Rules for Assigning Oxidation Numbers* to determine the oxidation number assigned to each element in each of the given chemical formulas.

	Formula	Element and Oxidation Number			
1.	Cl_2	Cl			
2.	Cl^-	Cl			
3.	Na	Na			
4.	Na^+	Na			
5.	O_2	O			
6.	N_2	N			
7.	Al^{+3}	Al			
8.	H_2O	H		O	
9.	NO_3^-	N		O	
10.	NO_2	N		O	
11.	$\text{Cr}_2\text{O}_7^{2-}$	Cr		O	
12.	KCl	K		Cl	
13.	NH_3	N		H	
14.	CaH_2	Ca		H	
15.	SO_4^{2-}	S		O	

	Formula	Element and Oxidation Number				
16.	Na_2O_2	Na		O		
17.	SiO_2	Si		O		
18.	CaCl_2	Ca		Cl		
19.	PO_4^{3-}	P		O		
20.	MnO_2	Mn		O		
21.	FeO	Fe		O		
22.	Fe_2O_3	Fe		O		
23.	H_2O_2	H		O		
24.	CaO	Ca		O		
25.	H_2S	H		S		
26.	H_2SO_4	H		S		O
27.	NH_4Cl	N		H		Cl
28.	K_3PO_4	K		P		O
29.	HNO_3	H		N		O
30.	KNO_2	K		N		O

Answer Key									
1.	Cl:0	7.	Al:3+	13.	N:3- H:1+	19.	P:5+ O:2-	25.	H:1+ S:2-
2.	Cl:1-	8.	H:1+ O:2-	14.	Ca:2+ H:1-	20.	Mn:4+ O:2-	26.	H:1+ S:6+ O:2-
3.	Na:0	9.	N:5+ O:2-	15.	S:6+ O:2-	21.	Fe:2+ O:2-	27.	N:3- H:1+ Cl:1-
4.	Na:1+	10.	N:4+ O:2-	16.	Na:1+ O:1-	22.	Fe:3+ O:2-	28.	K:1+ P:5+ O:2-
5.	O:0	11.	Cr:6+ O:2-	17.	Si:4+ O:2-	23.	H:1+ O:1-	29.	H:1+ N:5+ O:2-
6.	N:0	12.	K:1+ Cl:1-	18.	Ca:2+ Cl:1-	24.	Ca:2+ O:2-	30.	K:1+ N:3+ O:2-

Section 2: REDOX Reactions

Oxidized or Reduced?

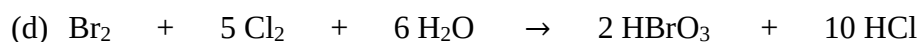
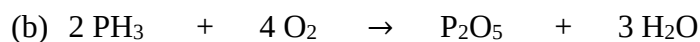
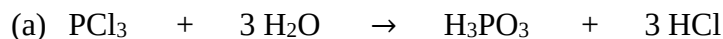
- In a redox reaction, electrons are transferred from one atom to another. One atom _____ electrons while another atom _____ electrons.
- This causes a _____ in the oxidation numbers of the affected atoms.
- Two great mnemonics!
 _____ says _____: A species **L**oses **E**lectrons when **O**xidized, and **G**ains **E**lectrons when **R**educed.
- _____ : Oxidation Is Loss (OIL) and Reduction Is Gain (RIG)



Oxidation and Reduction in Reactions

- Write a balanced, net-ionic equation for the reaction between zinc metal and aqueous hydrochloric acid. Assign oxidation states to all atoms.
- In the reaction that you wrote in part (b),
 zinc (loses gains) electrons, so it is (oxidized reduced). hydrogen (loses gains) electrons, so it is (oxidized reduced).

Practice! For each of the following reactions, classify them as redox equations (or not.) If a reaction is redox, assign oxidation numbers to each element on both sides of the equation. Identify the species that is oxidized and the species that is reduced.



Balancing Redox Equations

Red and Ox

- Redox reactions can be separated into _____ separate _____ REACTIONS- the oxidation process and the reduction process
- The pair of half reactions gives electrons (oxidation) and takes electrons (reduction) and therefore must be balanced both _____ AND _____ transferred.
- These half reactions can be taken out of the full equation to be examined

Balancing Half Reactions with Electrons

Balance for charge by adding the correct number of electrons to each

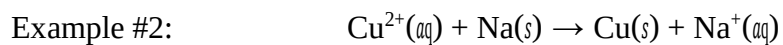
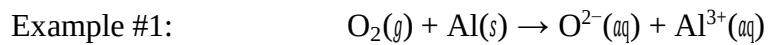
- $\text{NO}_2^- \rightarrow \text{NO}_3^-$
- $\text{MnO}_4^- \rightarrow \text{Mn}^{+2}$
- $\text{N}_2\text{O}_4 \rightarrow \text{NH}_3$

Label each as an oxidation or reduction. How can you tell?

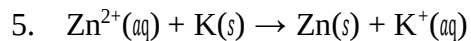
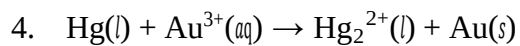
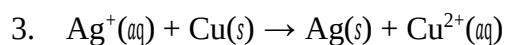
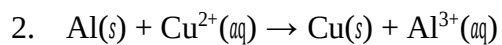
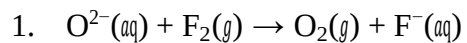
- The method of half reactions helps to ensure that the electrons gained (from a reduction half reaction) are _____ to the electrons lost (from an oxidation half reaction.)
- In the tables below, combine the two half reactions and write the overall balanced redox equation.

Reduction Half Reaction	Oxidation Half Reaction
$\text{Fe}^{3+} + e^- \rightarrow \text{Fe}^{2+}$	$\text{Sn}^{2+} \rightarrow \text{Sn}^{4+} + 2e^-$
Overall Redox Equation	

Reduction Half Reaction	Oxidation Half Reaction
$\text{NO}_3^- + 4\text{H}^+ + 3e^- \rightarrow \text{NO} + 2\text{H}_2\text{O}$	$5\text{H}_2\text{O} + \text{As}_2\text{O}_3 \rightarrow 2\text{H}_3\text{AsO}_4 + 4\text{H}^+ + 4e^-$
Overall Redox Equation	

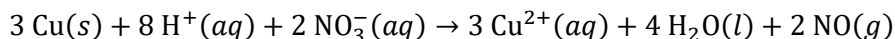
More Examples

Now You Try! Balance the following redox reactions.

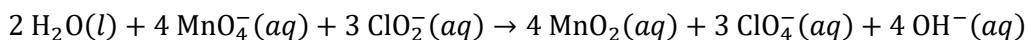


Multiple Choice Practice

1. A balanced equation for the reaction of copper metal with nitric acid is shown below. Which of the following represents a true statement about the reaction?



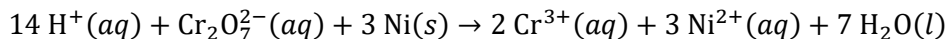
- The oxidation state of nitrogen changed from +5 to +2.
 - Hydrogen ions are oxidized to form $\text{H}_2\text{O}(l)$.
 - The oxidation state of oxygen changes from -1 to -2.
 - Copper metal is reduced to a copper (II) ion.
2. A strip of metal X is placed into a solution containing Y^{2+} ions and no reaction occurs. When metal X is placed in a separate solution containing Z^{2+} ions, metal Z starts to form on the strip. Which of the following choices organizes the reduction potentials for metals X, Y, and Z from greatest to least?
- $\text{X} > \text{Y} > \text{Z}$
 - $\text{Y} > \text{Z} > \text{X}$
 - $\text{Z} > \text{X} > \text{Y}$
 - $\text{Y} > \text{X} > \text{Z}$



3. Which species is reduced in the reaction represented above?

- MnO_2
- ClO_2^-
- MnO_4^-
- ClO_4^-

4. In the reaction below, a piece of solid nickel is added to a solution of potassium dichromate.



Which species is being oxidized and which is being reduced?

- | <u>Oxidized</u> | <u>Reduced</u> |
|-------------------------------------|----------------------------------|
| a. $\text{Cr}_2\text{O}_7^{2-}(aq)$ | $\text{Ni}(s)$ |
| b. $\text{Cr}^{3+}(aq)$ | $\text{Ni}^{2+}(aq)$ |
| c. $\text{Ni}(s)$ | $\text{Cr}_2\text{O}_7^{2-}(aq)$ |
| d. $\text{Ni}^{2+}(aq)$ | $\text{Cr}^{3+}(aq)$ |

Redox Titrations: Electrochemistry Stoichiometry in Action!

Important Caveat: at this point in time, we have only performed and analyzed _____ titrations, but a redox titration is _____ **AN ACID-BASE TITRATION**. Although much of our language and calculations will be the same, please be cognizant of the difference and DO NOT use acid-base terminology (especially in your lab report).

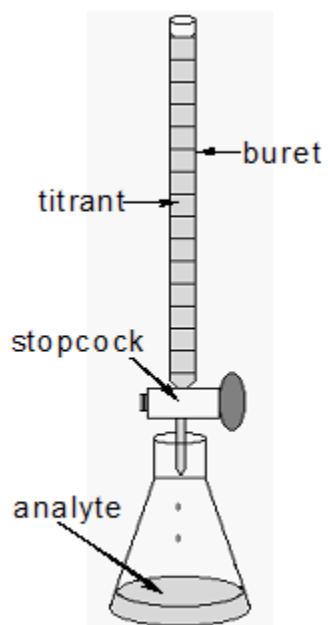
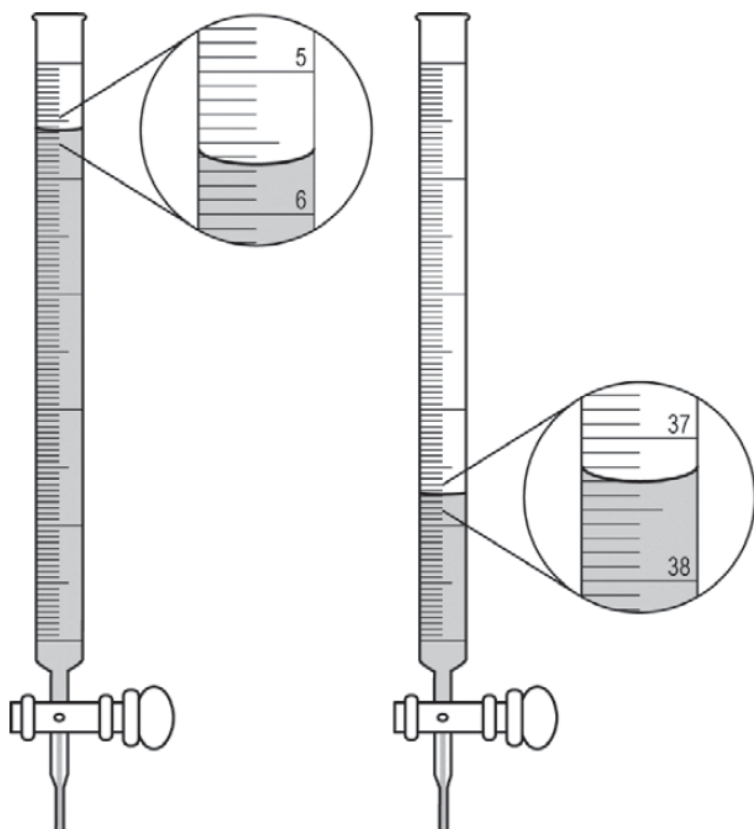


Figure 1: Titration Setup

Goal of Titration: determine the unknown concentration of the analyte!

A substance in a solution of _____ concentration (the _____, usually in a buret) is reacted with another substance in a solution of _____ concentration (the **analyte**, usually in a flask or beaker)

- **Equivalence point:** the point at which the moles of each reactant are **stoichiometrically equal** to each other in solution. (However, this is _____ where the two reactants have neutralized each other: that only happens in an acid-base titration!)
 - o **Equivalence point is _____ equilibrium!!!**
- **End point:** the point of the titration where an _____ changes color (if thoughtfully chosen, this occurs approximately at the equivalence point as a visual indication of the point at which the stoichiometric equivalent has been added).



Quick Reminder: How to Read a Buret

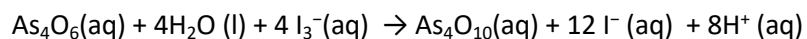
Burets, unlike most glassware used for precise volume measurements, are read from the _____ down, not the bottom up!

Example: The image to the left shows a buret of KMnO_4 before titration begins (leftmost) and at the end point (rightmost).

1. What volume of solution was initially present in the buret? _____
2. What volume was present at the end point of the titration? _____
3. How much KMnO_4 solution was added to the flask?

Let's Practice!

1. A solution of I_3^- (aq) can be standardized by using it to titrate As_4O_6 (aq) under acidic conditions. The titration of 0.1021 g of As_4O_6 (aq) dissolved in 30.00 mL of water requires 36.55 mL of I_3^- (aq) solution to reach equivalence point. The unbalanced equation is:



- Identify the species oxidized and reduced in this reaction.
- Calculate the moles of As_4O_6 (aq) present in the initial sample.
- Calculate the initial $[\text{As}_4\text{O}_6]$.
- How many moles of I_3^- (aq) were added to reach equivalence point?
- Calculate $[\text{I}_3^-]$.

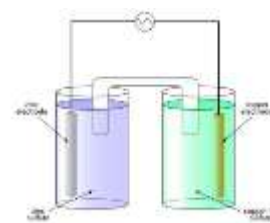
- When a strip of iron metal is placed in a solution of CuSO_4 , a reddish-brown solid appears on the surface of the iron. Write a balanced chemical equation for this reaction.

- In this reaction, iron is (reduced oxidized), and copper is (reduced oxidized).

- When a strip of copper metal is placed in a solution of FeSO_4 , a chemical reaction (does doesn't) take place. This is because iron (loses gains) electrons more readily than copper does. (Hint: Remember the Activity Series???)

- This means that forward reaction is spontaneous and the reverse is

nonspontaneous. Note: Which one has a $\Delta G < 0$?)



- In a voltaic (or galvanic) cell, the transfer of _____ takes place through an

_____ pathway rather than directly between reactants present in the same reaction vessel.

- Electrons flowing through a _____ and ions moving in solution both constitute an electrical current. This flow of electrical charge can be used to accomplish electrical work. (from the free energy!)
- The two solid metals connected by the external circuit are called _____.
- In a galvanic cell, oxidation always occurs at the _____, and reduction occurs at the _____.

A mnemonic device to help you remember this fact is

- “the vowels go together (O and A) and the consonants go together (R and C).”
- Another mnemonic is _____ - REDuction at the CAThode, ANode is OXidizing

- As the galvanic cell runs over time, the mass of the anode (decreases / increases) because the atoms of the metal are going into solution, becoming aqueous ions.
- As the galvanic cell runs over time, the mass of the cathode (decreases / increases) because the aqueous ions of the metal are converted into atoms and deposited on the surface of the cathode.

- The figure shows a zinc – copper galvanic cell.

The half-reaction that occurs at the anode is: _____

The half-reaction that occurs at the cathode is: _____

- In the diagram below, label the following.

anode

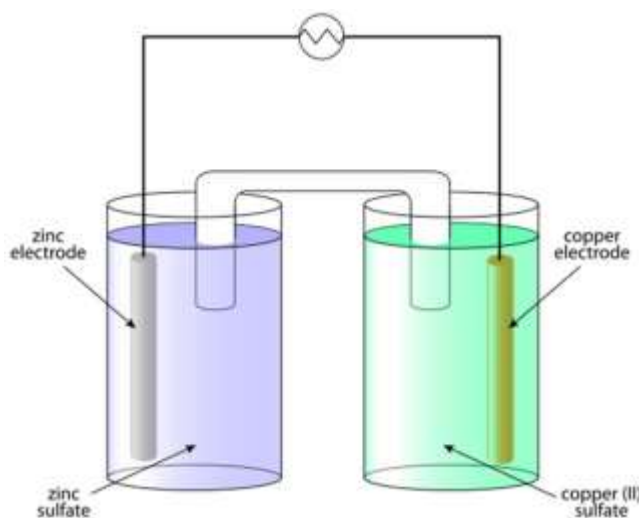
cathode

salt bridge

voltmeter.

In addition, draw an arrow to indicate the direction in which electrons are flowing through the wire.

_____ **do not flow through the salt bridge.** Only _____ flow through the salt bridge.



- For a galvanic cell to work, the solutions in the two half-cells must remain electrically _____. A porous glass disc or U-shaped tube acts as a _____ so that ions can travel through the electrolyte solution from one side to the other.
- The anions always migrate toward the (anode cathode) and the cations always migrate toward the (anode cathode).
- The electrons flow from the (anode cathode) toward the (anode cathode).

Electrons do NOT flow through the salt bridge! This is a common misconception.

Another mnemonic: _____: Electrons travel **F**rom **A**node **T**o **C**ATHode

Identify the anode and the cathode for each of the following galvanic cells.

Overall Reaction	Anode	Cathode
$2 \text{Ag}^+ + \text{Sn} \rightarrow 2 \text{Ag} + \text{Sn}^{2+}$		
$2 \text{Al} + 3 \text{Sn}^{2+} \rightarrow 2 \text{Al}^{3+} + 3 \text{Sn}$		
$\text{Cu}^{2+} + \text{Sn} \rightarrow \text{Cu} + \text{Sn}^{2+}$		
$2 \text{Ag}^+ + \text{Cu} \rightarrow 2 \text{Ag} + \text{Cu}^{2+}$		
$2 \text{Al} + 3 \text{Cu}^{2+} \rightarrow 2 \text{Al}^{3+} + 3 \text{Cu}$		

- Based on the information in part (o), which of the four metals (Ag, Sn, Al, or Cu) seems to lose electrons most easily? Which metal seems to gain electrons most easily?
- Only one of the following reactions actually occurs. Write a balanced net-ionic equation for the chemical reaction that does occur.

aluminum metal plus silver nitrate $\rightarrow$?

silver metal plus aluminum nitrate $\rightarrow$?

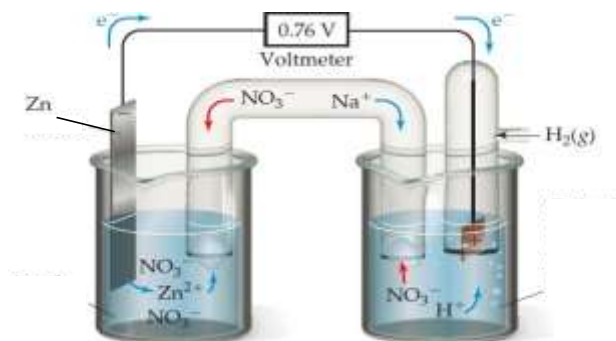
Section 4: Cell Potentials Under Standard Conditions

The reason that electrons flow from a Zn atom to a Cu^{2+} ion is based on a difference in potential energy. In general, which electrode (anode or cathode) has the *higher potential energy* in a galvanic cell?

- The potential energy of electrons is higher in the _____ than in the _____. Thus, the electrons always flow from the _____ to the _____.
- The difference in potential energy per electrical charge (the potential difference) between two electrodes is measured in units of _____.
- The potential difference between two electrodes of a galvanic cell is called the _____, denoted by E_{cell} . It is also called the electromotive force, or emf. It is also commonly called the _____ of the cell because it is measured in volts.

GALVANIC CELLS are ones with spontaneous electron flow!

- The cell potential of any galvanic cell is (positive negative).
 - (Remember, for thermo, spontaneous means _____ and for equilibrium, spontaneous means _____)
- The standard half-cell potentials (voltages) are tabulated under _____.
- Under standard conditions, the temperature is _____ °C, the pressure is _____ atm, and the concentration of reactants and products in solution is _____ mol/L.
- There are only tables for the reduction half cells, which means they denoted E_{red}° . Why aren't there tables for the oxidation half cells?
- The cell potential is calculated with the equation: $E_{cell}^{\circ} = E_{cat}^{\circ} - E_{an}^{\circ}$
- It is not possible to measure the standard reduction potential of a half reaction directly. The reference half reaction is the reduction of hydrogen $H^+(aq)$ to $H_2(g)$ under standard conditions, which is assigned a standard reduction potential of exactly _____ V. This is known as the Standard Hydrogen Electrode, or _____.



- Identify the half reactions that occur at each electrode in the diagram above.
Anode: _____ Cathode: _____
- In the galvanic cell shown above, the nitrate ions migrate toward the _____, and the sodium ions migrate toward the _____.
- If a potassium sulfate solution were used inside the salt bridge, the potassium ions would migrate toward the _____, and the sulfate ions would migrate toward the _____.

Cell potential and Stoichiometry..... hint.... There isn't any!!!

- In thermochemistry, if a substance in a balanced equation had a coefficient, we would multiply the value

of ΔH° or ΔS° or ΔG° by that coefficient for the sake of stoichiometry.

- E_{cell} _____ use that convention!!! Do NOT multiply by coefficients! Just use the formula

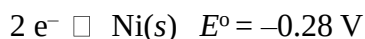
$$E_{\text{cell}}^\circ = E_{\text{cat}}^\circ - E_{\text{an}}^\circ$$

Half-Reaction	E° (V)
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	+0.80
$\text{Al}^{3+}(aq) + 3 e^- \rightarrow \text{Al}(s)$	-1.66
$\text{Cu}^{2+}(aq) + 2 e^- \rightarrow \text{Cu}(s)$	+0.34
$\text{Sn}^{2+}(aq) + 2 e^- \rightarrow \text{Sn}(s)$	-0.14

- Use the data in the table above to calculate the standard cell potential (E_{cell}°) for each of the following reactions.

Redox Reaction	E_{cell}° (V)
$2 \text{Ag}^+ + \text{Cu} \rightarrow 2 \text{Ag} + \text{Cu}^{2+}$	
$2 \text{Al} + 3 \text{Sn}^{2+} \rightarrow 2 \text{Al}^{3+} + 3 \text{Sn}$	

- Using the four half reactions listed in the table above, write the balanced net ionic equation for the redox reaction that would generate the largest value for E_{cell} .
- Determine the value of E_{cell}° for that redox reaction.
- A standard galvanic cell is set up using chromium and nickel. Based on the information below, determine which half-reaction occurs at the anode and which half-reaction occurs at the cathode. Calculate the standard cell potential for the cell.



Anode: _____

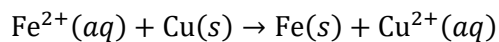
Cathode: _____

E_{cell}° : _____

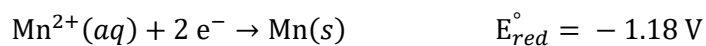
Multiple Choice Practice: Wheel!



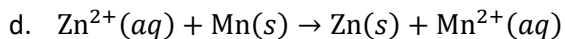
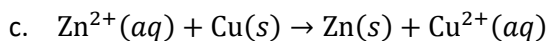
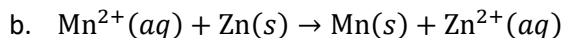
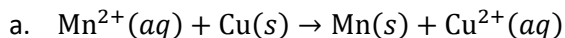
3. Based on the reduction potentials given above, what is the reaction potential for the following reaction?



- a. -0.78 V b. -0.10 V c. $+0.10 \text{ V}$ d. $+0.78 \text{ V}$



4. Based on the reduction potentials given above, which of the following reactions will be thermodynamically favored?



Thermodynamic Favorability and Redox Reactions

- Remember that the cell potential is calculated: $E_{cell}^0 = E_{cat}^0 - E_{an}^0$
- A positive value for the potential, E^0 , for a certain redox equation indicates that the process (is isn't) thermodynamically favored. A negative value for the potential, E^0 , for a certain redox equation indicates that the process (is isn't) thermodynamically favored.
- Use the table of standard reduction potentials to calculate the potential for each redox reaction and determine if the reaction is favored or not favored under standard conditions.

Redox Reaction	E^0 (V)	favored or not favored?
$3 \text{ Ag} + \text{Al}^{3+} \rightarrow 3 \text{ Ag}^+ + \text{Al}$		
$\text{Sn}^{2+} + 2 \text{ Fe}^{3+} \rightarrow \text{Sn}^{4+} + 2 \text{ Fe}^{2+}$		
$2 \text{ H}^+ + \text{Ca} \rightarrow \text{H}_2 + \text{Ca}^{2+}$		
$2 \text{ Na}^+ + 2 \text{ Br}^- \rightarrow 2 \text{ Na} + \text{Br}_2$		

Spontaneity (Thermodynamic Favorability) and Cell Potential

- In terms of thermochemistry, a reaction is thermodynamically favored if.....
- In terms of cell potential, at standard conditions
 - E_{cell}^0 ____ 0 in a thermodynamically favorable cell
 - E_{cell}^0 ____ 0 in a thermodynamically unfavorable cell
 - Therefore at standard conditions,
 - If $E_{cell}^0 > 0$, reaction is thermodynamically favorable, ΔG^0 ____ 0
 - If $E_{cell}^0 < 0$, reaction is not thermodynamically favorable, ΔG^0 ____ 0

Spontaneity (Thermodynamic Favorability) and Equilibrium

- In terms of equilibrium, a reaction is thermodynamically favored if.....
- In terms of equilibrium, at standard conditions
 - All concentrations are _____ M All gases are _____ atm
 - Therefore, at standard conditions, Q is ALWAYS = ____
- Recall: A forward reaction is _____ when _____
- Therefore at standard conditions,
 - If $K > 1$, reaction is thermodynamically _____, ΔG^0 ____ 0
 - If $K < 1$, reaction is thermodynamically _____, ΔG^0 ____ 0

SPONTANEOUS REACTIONS have:	NONSPONTANEOUS REACTIONS have:

- This means that ΔG , E° and K are all _____

Thermodynamic Favorability can be Calculated

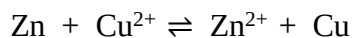
- The thermodynamic favorability of an electrochemical cell can be calculated with the equation:

$$\Delta G = -nFE^\circ$$

- In this equation, n is a positive number without units that represents the number of _____ of _____ transferred according to the balanced equation for the reaction.
- F is Faradays Constant and is equal to 96,485 J/m electron transferred
- Recall: Thermodynamic Favorability is related to K value and can be calculated with the equation:

$$\bullet \quad \Delta G = -RT \ln K$$

- Calculate the value of E° , ΔG° (in units of kJ/mol_{rxn}) and K for the following reaction under standard conditions.



Practice

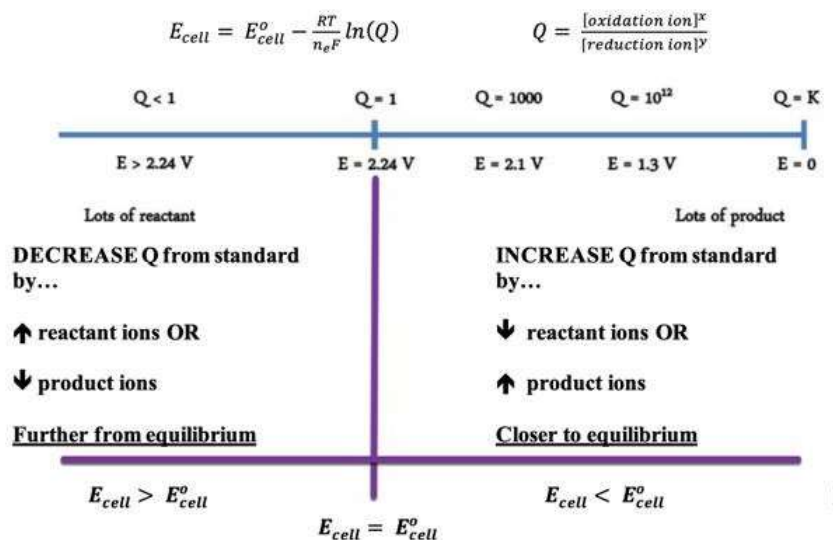
- For the two half cells: $\text{Fe}^{+2} + 2\text{e}^- \rightarrow \text{Fe}$ and $\text{Cu}^{+1} + 1\text{e}^- \rightarrow \text{Cu}$
 - Write an overall balanced equation
 - Calculate E_{cell}
 - Calculate ΔG
 - Calculate K
 - Calculate Q for a standard cell

Section 6 – Cell Potentials Under Nonstandard Conditions

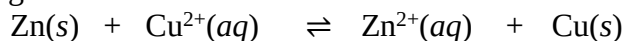
Standard vs Nonstandard

- Recall that the standard reduction potentials of a cell are tabulated at standard conditions. Under standard conditions, the temperature is _____ $^{\circ}\text{C}$, the pressure is _____atm, and the concentration of all solutions _____ mol/L.
- Calculate the reaction quotient for the $\text{Zn} + \text{Cu}^{2+} \rightleftharpoons \text{Zn}^{2+} + \text{Cu}$ cell:
 - The reaction quotient or Q value for **any** cell at standard conditions is _____. For most cells, this Q value is much (less than greater than) K , and results in a highly spontaneous reaction.
 - As a galvanic cell is discharged, the concentrations of the ions in solution, originally at 1.0 M in each half-cell, will change gradually over time. As the cell runs, the concentration of the reactants (*decreases increases*), and the concentration of the products (*decreases increases*) and therefore the Q value (*increases decreases*), and gets (*closer to farther from*) K .
 - Because of this shift in Q , as the cell runs, the _____ progressively (*decreases increases*) over time. (the cell gets weaker!)
 - When the cell reaches equilibrium, the value of E becomes equal to _____V, and the value of ΔG becomes equal to _____kJ

ENE-6.C.2 Equilibrium arguments such as Le Châtelier's principle do not apply to electrochemical systems, because the systems are not in equilibrium.



For example, consider the following reaction



Under standard conditions, the galvanic cell has the following properties. Note that the reaction is favored (negative ΔG) and the equilibrium constant K is very large (i.e., product-favored).

[Cu ²⁺] (M)	[Zn ²⁺] (M)	Q	E (V)	ΔG (kJ/mol _{rxn})	K
1.0	1.0	1.0	1.10	-212	1.5 x 10 ³⁷

When this galvanic cell reaches equilibrium, $Q = K$, and the cell has the following properties.

[Cu ²⁺] (M)	[Zn ²⁺] (M)	Q	E (V)	ΔG (kJ/mol _{rxn})
very small	very large			

The following tables illustrate the effect of changes in concentration (of reactants or products) on the voltage (E) of the galvanic cell. For each, calculate Q , and indicate if E_{cell} is greater than, less than or equal to E°_{cell} , and if ΔG is greater than, less than or equal to ΔG° .

[Cu ²⁺] (M)	[Zn ²⁺] (M)	Q	E (V)	ΔG (kJ/mol _{rxn})
10.0	1.0			

Decreasing the Concentration of the Products: $Q < 1$ and $E > E^{\circ}$

[Cu ²⁺] (M)	[Zn ²⁺] (M)	Q	E (V)	ΔG (kJ/mol _{rxn})
1.0	0.010			

Increasing the Concentration of the Products: $Q > 1$ and $E < E^\circ$

$[\text{Cu}^{2+}]$ (M)	$[\text{Zn}^{2+}]$ (M)	Q	E (V)	ΔG (kJ/mol _{rxn})
1.0	10.0			

Decreasing the Concentration of the Reactants: $Q > 1$ and $E < E^\circ$

$[\text{Cu}^{2+}]$ (M)	$[\text{Zn}^{2+}]$ (M)	Q	E (V)	ΔG (kJ/mol _{rxn})
0.010	1.0			

Summary of Cell Potentials Under Nonstandard Conditions

Increasing the concentration of reactants or decreasing the concentration of products increases the driving force for the reaction, resulting in a voltage that is GREATER than E° .

If $Q < 1$, you are FURTHER AWAY FROM EQUILIBRIUM than $Q = 1$ and E is GREATER than E° .

Decreasing the concentration of reactants or increasing the concentration of products decreases the driving force for the reaction, resulting in a voltage that is LESS than E° .

If $Q > 1$, you are CLOSER TO EQUILIBRIUM than $Q = 1$ and E is LESS than E° .

The value of E under nonstandard conditions can be calculated with the **Nernst equation**

The AP Chemistry exam does not require students to use the Nernst equation to do calculations involving E , and the Nernst equation is not included on the list of equations and constants given to students on the exam. **DO NOT USE IT IF YOU CANNOT EXPLAIN THE CHEMISTRY IT REPRESENTS.**

The following information comes from the AP Chemistry Course and Exam Description.

Many real systems do not operate at standard conditions and the electrical potential determination must account for the effect of concentrations.

The qualitative effects of concentration on the cell potential can be understood by considering the cell potential as a driving force toward equilibrium, in that the farther the reaction is from equilibrium, the greater the magnitude of the cell potential.

The standard cell potential, E° , corresponds to the standard conditions of $Q = 1$.

As the system approaches equilibrium, the magnitude (i.e., absolute value) of the cell potential decreases, reaching zero at equilibrium (when $Q = K$).

Deviations from standard conditions that take the cell further from equilibrium than $Q = 1$ will increase the magnitude of the cell potential relative to E° .

Deviations from standard conditions that take the cell closer to equilibrium than $Q = 1$ will decrease the magnitude of the cell potential relative to E° .

In concentration cells, the direction of spontaneous electron flow can be determined by

Section 7– Electrolytic Cells

- Galvanic cells are based on spontaneous (favorable) redox reactions. It is also possible for nonspontaneous redox reactions to occur by using _____ to larger energy sources to drive them. These are known as _____ cells.
- The reverse reaction of a _____ cell is its equal and opposite _____ cell.
- One application of electrolytic cells is the recapture of _____ through _____. (What kind of reaction did we learn this as originally?)
- Write the reaction for the electrolysis of molten sodium chloride:
- Write the balanced half reactions for the electrolysis of molten sodium chloride.
Reduction at the cathode: _____
Oxidation at the anode: _____
- The process of _____ uses electrolysis to deposit a thin layer of one metal on another metal to improve beauty or resistance to corrosion.
- The quantity of charge passing through an electrical circuit is measured in units of _____. This unit is abbreviated as C, and it can be calculated by multiplying (amperes) x (seconds). Note that 1 ampere = 1 C/s.

Ex. How many grams of chromium can be plated out by passing 2.05 amps through acidic CrO_3 for 1.0×10^4 s?

Ex. How many hours would it take to produce 25.0 g of Cr from a solution of CrCl_3 by a current of 2.75A?

Suppose that molten AlCl_3 undergoes electrolysis with a current of 10.0 A.

- (i) How many grams of aluminum would be produced in one hour?
- (ii) How many minutes would it take for 1.75 g Al to be produced?

TABLE 18.1 Standard Electrode Potentials at 25 °C

Reduction Half-Reaction			E° (V)
Stronger oxidizing agent 	$F_2(g) + 2 e^-$	$\longrightarrow 2 F^-(aq)$	2.87
	$H_2O_2(aq) + 2 H^+(aq) + 2 e^-$	$\longrightarrow 2 H_2O(l)$	1.78
	$PbO_2(s) + 4 H^+(aq) + SO_4^{2-}(aq) + 2 e^-$	$\longrightarrow PbSO_4(s) + 2 H_2O(l)$	1.69
	$MnO_4^-(aq) + 4 H^+(aq) + 3 e^-$	$\longrightarrow MnO_2(s) + 2 H_2O(l)$	1.68
	$MnO_4^-(aq) + 8 H^+(aq) + 5 e^-$	$\longrightarrow Mn^{2+}(aq) + 4 H_2O(l)$	1.51
	$Au^{3+}(aq) + 3 e^-$	$\longrightarrow Au(s)$	1.50
	$PbO_2(s) + 4 H^+(aq) + 2 e^-$	$\longrightarrow Pb^{2+}(aq) + 2 H_2O(l)$	1.46
	$Cl_2(g) + 2 e^-$	$\longrightarrow 2 Cl^-(aq)$	1.36
	$Cr_2O_7^{2-}(aq) + 14 H^+(aq) + 6 e^-$	$\longrightarrow 2 Cr^{3+}(aq) + 7 H_2O(l)$	1.33
	$O_2(g) + 4 H^+(aq) + 4 e^-$	$\longrightarrow 2 H_2O(l)$	1.23
	$MnO_2(s) + 4 H^+(aq) + 2 e^-$	$\longrightarrow Mn^{2+}(aq) + 2 H_2O(l)$	1.21
	$IO_3^-(aq) + 6 H^+(aq) + 5 e^-$	$\longrightarrow \frac{1}{2} I_2(aq) + 3 H_2O(l)$	1.20
	$Br_2(l) + 2 e^-$	$\longrightarrow 2 Br^-(aq)$	1.09
	$VO_2^+(aq) + 2 H^+(aq) + e^-$	$\longrightarrow VO^{2+}(aq) + H_2O(l)$	1.00
	$NO_3^-(aq) + 4 H^+(aq) + 3 e^-$	$\longrightarrow NO(g) + 2 H_2O(l)$	0.96
	$ClO_2(g) + e^-$	$\longrightarrow ClO_2^-(aq)$	0.95
	$Ag^+(aq) + e^-$	$\longrightarrow Ag(s)$	0.80
	$Fe^{3+}(aq) + e^-$	$\longrightarrow Fe^{2+}(aq)$	0.77
	$O_2(g) + 2 H^+(aq) + 2 e^-$	$\longrightarrow H_2O_2(aq)$	0.70
	$MnO_4^-(aq) + e^-$	$\longrightarrow MnO_4^{2-}(aq)$	0.56
	$I_2(s) + 2 e^-$	$\longrightarrow 2 I^-(aq)$	0.54
	$Cu^+(aq) + e^-$	$\longrightarrow Cu(s)$	0.52
	$O_2(g) + 2 H_2O(l) + 4 e^-$	$\longrightarrow 4 OH^-(aq)$	0.40
	$Cu^{2+}(aq) + 2 e^-$	$\longrightarrow Cu(s)$	0.34
	$SO_4^{2-}(aq) + 4 H^+(aq) + 2 e^-$	$\longrightarrow H_2SO_3(aq) + H_2O(l)$	0.20
	$Cu^{2+}(aq) + e^-$	$\longrightarrow Cu^+(aq)$	0.16
	$Sn^{4+}(aq) + 2 e^-$	$\longrightarrow Sn^{2+}(aq)$	0.15
	$2 H^+(aq) + 2 e^-$	$\longrightarrow H_2(g)$	0
	$Fe^{3+}(aq) + 3 e^-$	$\longrightarrow Fe(s)$	-0.036
	$Pb^{2+}(aq) + 2 e^-$	$\longrightarrow Pb(s)$	-0.13
	$Sn^{2+}(aq) + 2 e^-$	$\longrightarrow Sn(s)$	-0.14
	$Ni^{2+}(aq) + 2 e^-$	$\longrightarrow Ni(s)$	-0.23
	$Cd^{2+}(aq) + 2 e^-$	$\longrightarrow Cd(s)$	-0.40
	$Fe^{2+}(aq) + 2 e^-$	$\longrightarrow Fe(s)$	-0.45
	$Cr^{3+}(aq) + e^-$	$\longrightarrow Cr^{2+}(aq)$	-0.50
	$Cr^{3+}(aq) + 3 e^-$	$\longrightarrow Cr(s)$	-0.73
	$Zn^{2+}(aq) + 2 e^-$	$\longrightarrow Zn(s)$	-0.76
	$2 H_2O(l) + 2 e^-$	$\longrightarrow H_2(g) + 2 OH^-(aq)$	-0.83
	$Mn^{2+}(aq) + 2 e^-$	$\longrightarrow Mn(s)$	-1.18
	$Al^{3+}(aq) + 3 e^-$	$\longrightarrow Al(s)$	-1.66
	$Mg^{2+}(aq) + 2 e^-$	$\longrightarrow Mg(s)$	-2.37
	$Na^+(aq) + e^-$	$\longrightarrow Na(s)$	-2.71
	$Ca^{2+}(aq) + 2 e^-$	$\longrightarrow Ca(s)$	-2.76
	$Ba^{2+}(aq) + 2 e^-$	$\longrightarrow Ba(s)$	-2.90
Weaker oxidizing agent 	$K^+(aq) + e^-$	$\longrightarrow K(s)$	-2.92
	$Li^+(aq) + e^-$	$\longrightarrow Li(s)$	-3.04
Weaker reducing agent			Stronger reducing agent